MILITARY SPECIFICATION

PROJECTILE, 155MM, GB2, M687, METAL PARTS ASSEMBLY WITH LOADED OGIVE

This specification is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains the requirements, quality assurance provisions and packaging for the Metal Parts Assembly with Loaded Ogive of Projectile, 155MM, M687.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. (see 6.2) Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-O-60371	-Oxamide Explosive Composition

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410	-Nondestructive Testing Personnel Qualification and Certification
MIL-STD-00453	-Inspection, Radiograph

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSR N/A FSC 1320

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGING DRAWINGS

11751156	- Projectile, 155MM, GB2, M687 Ogive
9357729	- Projectile, 155MM, GB2, M687 Marking Diagram
9349367	- Projectile, 155MM, GB2, M687, Metal Parts Assembly with Loaded Ogive

US ARMY CHEMICAL RESEARCH AND DEVELOPMENT CENTER (CRDC)

PRODUCT AND PACKAGING DRAWINGS

E15-12-321 -Pallet, Side Loading for 155MM, M687 Projectile

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Loaded ogive. The loaded ogive shall comply with all the requirements specified on dwg. 9349367 and meet the requirements specified in applicable specifications.

3.3 Clean explosive scrap. Clean explosive scrap resulting from normal production operations may be re-melted and used provided that it complies with the requirements for high explosive (including grade) as specified on dwg. 9349367 with the exception that it does not have to meet the material form requirement called for in MIL-O-60371. Scraps from the floor, ventilation and suction apparatus (other than from machining operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.8).

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3.4 Cavitation in high explosive charge.

3.4.1 General. Cracks and cavities within an explosive charge shall not exceed the requirements specified in Table I when determined as specified in 4.5.1. Cavities having a maximum (max) projected length of 1/16 inch shall be disregarded provided that they are separated by 1/2 inch or more of solid explosive.

TABLE I. Cavitation requirements

	A	B	C*
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings, square inch (sq. in.).	1/4	1/2	NA
Projected length of any cavity, excluding pipes, cracks, and annular rings (in.).	1/2	1/2	NA
Piping cavities, maximum (max.) projected area (sq. in.).	1/8	1/4	NA
Piping cavities, max. projected width (in.)	1/8	1/4	NA
Cracks, max. projected width (in.).	1/32	1/32	NA
Annular rings, max. projected width (in.).	0	1/16	NA

(Segments, See Figure 1)

Piping cavitation is defined as cavitation located on or near the vertical center line of the ogive with the longitudinal axis (length) at least three times or greater than the width.

An annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast.

* Void permitted in Segment C.

3.4.2 Cracked charges. There shall be no more than three transverse cracks in any total charge, and not more than one traverse crack shall be permitted in Segment A.

3.4.3 Porosity. Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

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3.4.4 Base separation. The maximum allowable base separation between the base of the ogive cavity and the cast shall be .015 of an inch when determined in accordance with 4.5.3.

3.4.5 Surface cavitation. Voids or cracks in segment C of Figure 1 shall not be cause for rejection. Cavities in the base of the fuzewell (top of segment B) shall not be cause for rejection provided the sum of the areas of the cavities is not greater than 80 percent of the top of segment B. No point may extend above the flat surface on which the liner will rest. Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in surface cavitation computations.

3.5 Loading of high explosive.

3.5.1 Loading. The ogive shall be loaded in accordance with dwg. 9349367.

3.5.2 Mating surfaces. The mating surfaces of threads shall be free of caked explosives and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the ogive.

3.5.3 Solid explosives. Solid explosives shall not be in the ogive before loading or added to the ogive during initial pouring or cooling.

3.5.4 Ogive interior cavity. The interior cavity of the ogive prior to loading shall be free from foreign matter.

3.5.5 Fuze well. The fuze well including threads shall be free from any foreign matter or loose explosive prior to assembly of the liner.

3.5.6 Exterior surfaces. There shall be no explosive on any exterior surfaces of the projectile after loading.

3.5.7 Ogive cavity. The ogive cavity shall be free of scales, fins, burrs, draw marks, laminations, embedded foreign matter, folds, pits and sharp edges.

3.6 Lifting plug. The projectile metal parts assembly with loaded ogive and with lifting plug assembled shall meet the disassembly torques specified on dwg. 9349367.

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3.7 Proving ground.

3.7.1 Metal parts assembly with inert ogive*. There shall be no evidence of metal parts failure when tested at excess pressure. Metal parts failure shall be:

a. Loss or breakup of metal part components in bore or in flight (except obturator).
b. Shear of the rotating band as evidenced by complete lack of engraving.

* Applicable only when ballistic test is used to qualify both metal parts and ogive loading.

3.7.2 Metal parts assembly with loaded ogive. There shall be no evidence of metal parts assembly with loaded ogive failure when tested for excess pressure or at zone 3. Failure of this configuration shall be:

a. Premature burst of the oxamide explosive composition in bore or in flight.
b. Partial or incomplete functioning upon impact.
c. Any metal parts separation in bore or in flight.

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship. The requirements of workmanship are as specified on the applicable drawings, reference specifications and the following:

3.9.1 Cleaning. If required, the cleaning method shall not be injurious to any of the part(s) nor shall the parts be contaminated by the cleaning agent.

3.9.2 Protection of the rotating band. Care shall be exercised at all times so that the rotating band of each projectile will not be nicked or burred.

3.9.3 Foreign matter. No part or assembly shall contain dirt, grease, chips, oil, rust corrosion or other foreign matter prior to inspection and shipping.

3.9.4 Obturator. Care shall be taken at all times to protect obturator from being damaged.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Projectile Metal Parts Assembly complete with Base-Adapter and Loaded Ogives	9349367	20
Loaded Ogives	9349367	8

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-64005 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2	Metal Parts Assembly with Loaded Ogive				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Loaded Ogives (Dwg. 9349367)				
	Base separation of loaded ogive	8		3.4.4	4.4.3.1.1
	Cavitation check of loaded ogive	8		3.4	4.5.1.1 to 4.5.1.4 except that any convenient orientation may be used for radiograph of sectioned loaded ogive.
	Special Radiographic inspection of sectioned loaded ogive. After the test for base separation, each sample shall be completely sectioned and each half radiographed for the purpose of detailed examination of the explosive filler.				
	Metal parts assembly with loaded ogive (Dwg. 9349367)				
	Examination for defects	20		3.2	4.4.2.2 to 4.4.2.4, 4.4.2.6 (NOTE 2)
	Ballistic tests:				
	A. Zone 3 (-40 + 5°F)	5		3.7.2	4.5.2.2
	B. Excess Pressure (+70°F)	15		3.7.2	4.5.2.3

NOTES: 1. With the exception of ballistic tests first article tests shall be done at the loading facility.
2. For these inspections rounds will not be disassembled-- only external tests will be done.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Metal Parts Assembly with loaded ogive shall contain:

a. Projectile metal parts of one lot interfix number from one manufacturer.
b. Oxamide explosive composition of the same type from one lot interfix number.
c. Fuze well liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.2.2 Examinations. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64005 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.1	Projectile Subassembly (prior to ogive loading)		1 of 1	11751156
				NEXT HIGHER ASSEMBLY: 9349367

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Foreign matter in ogive cavity		100%	3.5.4	Visual
2	Interior metal defects		100%	3.5.7	Visual
Major					
101	Solid explosive in ogive cavity		100%	3.5.3	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64005 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Metal Parts Assy with Loaded Ogive (w/o liner)	1 OF 1	9349367
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Presence of explosive on exterior of metal parts assy		100%	3.5.6	Visual
Major					
101	Depth of fuze well		.40%	3.2	Gage
102	Foreign matter in fuze well		.40%	3.5.5	Visual
103	Explosive in threads		100%	3.2	Visual
Minor					
201	Surface cavitation excessive		.65%	3.4.5	Visual
202	Diameter of fuze well, max.		.65%	3.2	Gage
203	Evidence of poor workmanship		.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64005 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.3	Metal Parts Assy with Loaded Ogive (with liner)		1 of 1	9349367
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH / PARAGRAPH REFERENCE /INSPECTION METHOD
Critical				
1	Liner missing		100%	3.2 Visual
Major				
101	True position of inside diameter of liner with pitch diameter of thread and front face of ogive.		.40%	3.2 Gage
102	Depth to bottom liner, min.		.40%	3.2 Gage
103	Inside diameter of liner, min. (entire length)		100%	3.2 Gage
Minor				
201	Liner damaged		.65%	3.2 Visual
202	Explosive or foreign matter in or on top of liner cavity		.65%	3.2 Visual
203	Liner can be twisted or pulled out by hand		.65%	3.2 Manual
204	Security of liner in cavity and against nose threads		.65%	3.2 Manual
205	Evidence of poor workmanship		.65%	3.9 Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64005 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Metal Parts Assy with Loaded Ogive (incl. lifting plug)	OF 1	9349367/9357729
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of bourrelet, max. (applies if bourrelet has been retouched)		100%	3.2	Gage
102	Lifting plug gasket missing		0.40%	3.2	Visual
103	Disassembly torque test of lifting plug		0.40%	3.6	4.4.5
104	Broken band square marking missing, incorrect or unidentifiable		100%	3.2	Visual
105	Weight zone marking missing, incorrect or unidentifiable		100%	3.2	Visual
106	Yellow band missing or incorrect		100%	3.2	Visual
107	Metal stamp letters missing, incorrect loose		.40%	3.2	Visual
108	Marking cover missing, incorrect or loose		100%	3.2	Visual
109	Rotating band damaged		100%	3.9.2	Visual
Minor					
201	Grommet missing, damaged or improperly assembled		0.65%	3.2	Visual
202	Obturator missing or damaged		100%	3.9.4	Visual
203	Marking missing, unidentifiable or incorrect color code		100%	3.2	Visual
204	Protective coating damaged		0.65%	3.9	Visual
205	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64005 (AR)

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.5	Metal Parts Assy with Loaded Ogive		1 1	9349367
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH / PARAGRAPH REFERENCE /INSPECTION METHOD
Critical I	Ballistic test			4.5.2
	a. Excess pressure; inert ogive*	3		3.7.1 4.5.2.1
	b. Zone 3 (-40 + 5°F); loaded ogive	5		3.7.2 4.5.2.2
	c. Excess pressure (+ 70°F); loaded ogive	10		3.7.2 4.5.2.3

NOTE: *Applicable only when metal parts are manufactured at the same facility as the loading of the ogive and the ballistic test is used to qualify both the metal parts and the ogive loading.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-64005 (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.6	Pallet		1	1	E15-12-321
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	None				
Minor					
201	Pallet damaged (unserviceable)		.65%	3.1	Visual
202	Improperly packed		.65%	3.1	Visual
203	Bindings missing, loose or broken		.65%	3.1	Visual/Manual
204	Marking missing, incorrect or illegible		.65%	3.1	Visual
205	Evidence of poor workmanship		.65%	3.9	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables and the following:

4.4.3.1 Cavitation (see 3.4, except 3.4.5). Cavitation defects are classified as follows:

Critical:

1. Cavitation, cracks, piping, porosity or annular rings in segment A or B in excess of that permitted in Table I.
2. Cracks in excess of that permitted by 3.4.2.
3. Base separation (see 3.4.4).

This test shall be performed on all loaded ogive assemblies (100 percent inspection). The radiographic test shall be performed as specified in 4.5.1 using equipment in accordance with 4.4.4. Any ogive which has any one of the above critical defects shall be classed defective and removed from the lot.

4.4.3.1.1 Base separation. The base separation inspection shall be performed either by sectioning of selected ogives (4.4.3.1.1.1) or by use of a radiographic standard (4.4.3.1.1.2).

4.4.3.1.1.1 Ogive sectioning. Ogives showing the largest base separation as determined by radiographic examination shall be selected for sectioning at the following rate. From every group of 700 consecutively produced ogives or portion thereof one selectioning sample shall be selected. If the sample fails, the requirement of 3.4.4, then all ogives in that group which show any indication of base separation by radiographic examination shall be rejected.

If radiographic examination of an entire group reveals no ogives with base separation, a sectioning sample shall be randomly selected. If the randomly selected sample fails to meet the requirement of 3.4.4, the entire group of ogives represented by that sample shall be rejected.

The sectioning samples shall be tested in accordance with 4.5.3.

4.4.3.1.1.2 Radiographic standard for base separation. A radiographic standard for base separation shall be developed by the contractor and approved by AMSMC-QAR-Q(D) prior to its use as a standard for acceptance. The standard shall utilize the bottom end of the ogive, cut approximately 1.25 inches above the ogive shoulder. A removable plug of epoxy shall be cast in the cut-off ogive. Radiographs at simulated base separations of .010, .015 and .020 inch shall be taken in order to brakcet the simulated

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base separation of .015 inch for sensitivity. The radiographic standard shall be verified at a minimum by the daily radiograph of the .015 inch simulated base separation cut-off ogive. Acceptance of loaded ogives shall be made by comparison of the sample radiograph to the standard.

4.4.3.2 Proving ground. The test shall be conducted in accordance with 4.5.2. If any projectile assembly fails to comply with the applicable requirement, the lot shall be rejected.

4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein, the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification and calibration checks during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of item produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

4.4.4.2 NDT personnel qualification and certification requirements. For setting up and performing tests using x-ray equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I or higher. Level I personnel shall work under the technical supervision of Level II personnel. For set-up and calibration of x-ray equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II. Personnel who read and interpret radiographic images and prepare radiographic inspection reports shall be certified to Level II or higher. A person certified to Level III shall approve all radiographic procedures for exposure, interpretation, reporting and changes thereto and be available for resolution of radiographic problems.

4.4.5 Lifting plug. With the lifting plug assembled to the projectile metal parts assembly, the disassembly torque shall not exceed 200 inch/pounds nor the continuous loosening torque be less than 100 inch/pounds. Any unit that fails to meet either of the torques required shall be classed defective and removed from the lot.

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4.5 Methods of inspection.

4.5.1 Radiographic examination.

4.5.1.1 Determination of cavities. In accordance with 4.4.3.1, projectile metal parts assemblies with loaded ogives shall be subjected to radiographic examination. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the projectile and tangent to the base of the ogive's cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks or other extraneous features shall project into the region of explosive under inspection with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the ogive. Such screening should be free of any radiographically observable feature in the projected area under inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-00453 (see 6.6). In addition, the sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of the explosive with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.5.1.2 and 4.5.1.3. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless identified as extraneous artifacts or metal parts irregularities. In the event an original radiograph leaves doubt as to the ogive's acceptability (e.g. difficulty in evaluating size, location or classification of a defect), the projectile assembly shall be rotated about its longitudinal axis from the original exposure by 90 degrees and re-radiographed for clarification. Acceptability shall be judged on the basis of the film giving the largest projection of the defect.

4.5.1.2 Check of sensitivity of radiographic technique. The sensitivity of the radiographic technique shall be determined by means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the ogive away from the film (source side). The penetrameter shall be positioned with the 2T hole superimposed on the ogive axis and tangent to the interior base, as shown in Figure 1. Observation shall be made of the image of the penetrameter on the radiograph film. The sensitivity shall be considered satisfactory when the image of the

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penetrameter, including the 2T hole, is sharply defined on the film. The penetrameter shall be made of steel having a specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and aluminum thickness and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole, the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter, see MIL-STD-00453.

4.5.1.3 Photographic density of radiograhic film. The photographic density of the film of ogives shall be determined in the region of minimum film density within the projected area of the explosive charge and shall measure 1.25 or greater. Film density shall be checked each hour of operation or fraction thereof for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units and having a sample spot of 3 millimeters diameter. Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurements on the radiographic film. Calibration shall be made at a step tablet density level within .2D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) or no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARDC.

Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet used in daily operations having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.1.4 Radiographic film. Only a film type and vendor as listed herein is approved for use in radiographing this loaded ogive. The final choice of film type must be determined for the particular radiographic facility to be used and will be based on achieving adequate sensitivity on both standard and penetrameters test items. A substitute item shall not be used before prior testing and approved by ARDC, Dover, New Jersey 07801-5001.

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Eastman Kodak Company
1187 Ridge Road, West
Rochester, New York 14650
Film Types: AA, AX, M, T, TMX

S.I. DuPont deNemours and Company
Photo Products Department
Willmington, Delaware 19898
Film Types: NDT-55. NDT-65, NDT-70

GAF Corporation
140 West 51st Street
New York, New York 10020
Film Types: 400

Processing chemicals and conditions shall be as recommended by film manufacturer.

4.5.2 Proving ground ballistic tests. The following tests shall be conducted using cannon tubes which have not reached the fourth quarter of remaining life based upon current gun tube condemnation criteria. Photographic equipment shall be positioned near the muzzle of the weapon to record the condition of the projectile as it exits the tube. Radar equipment shall be used to tract projectile flight. Chamber pressures and velocities shall be recorded for each round fired. All photographs and recovered hardware shall be examined for any noncompliance with the applicable requirements in 3.7. Range, deflection, and velocity shall be recorded for each round fired for information. An attempt will be made to recover the inert projectiles fired for this test.

4.5.2.1 Excess pressure (inert ogive).* Each projectile submitted for this test shall contain a simulated payload (canisters with OPA and DF simulant, N-Propyl Bromide, GFM) with its base assembled, an inert explosive charge, and a dummy fuze. Each projectile will be fired in a M199 Cannon with a M203 Propelling Charge conditioned to 70°F. Pressure/time traces shall be recorded (using instrumentation with piezoelectric crystal elements) to determine if chamber pressures are approaching or exceeding 47,000 psi. In addition, thermocouple measurements of charge temperature shall be taken during preconditioning and just prior to firing. An attempt will be made to recover the projectiles fired for this test.

* Applicable only when metal parts are manufactured at the same facility as the loading of the ogives and the acceptance tests are run concurrently.

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4.5.2.2 Zone 3. Each projectile submitted for this test shall contain a simulated payload (canisters with OPA and DF simulant, N-Propyl Bromide, GFM) with its base assembled, an ogive loaded with Comp B oxamide explosive and a M557 fuze. Each projectile shall be fired in either a M185 or M199 Cannon with a M3A1 Propelling Charge conditioned to -40 + 5°F.

4.5.2.3 Excess pressure (loaded ogive). Each projectile submitted for this test shall contain a simulated payload (canisters with OPA and DF simulant, N-Propyl Bromide, GFM) have its base assembled, the ogive loaded with Comp B- Oxamide explosive and a M557 fuze. Each projectile shall be fired in a M199 Cannon with an M203 charge conditioned to 70°F. Pressure/time traces shall be recorded (using instrumentation with piezoelectric crystal elements) to determine if chamber pressures are approaching or exceeding 47,000 psi. In addition, thermocouple measurements shall be taken during preconditioning and just prior to firing.

4.5.3 Determination of base separation. The sample(s) selected (see 4.4.3.1.1.1) shall be sectioned in a manner to expose the cast and the interior of the ogive at approximately the center of the base. Care shall be taken not to leave a burr on the interior of the ogive base. The ogive shall be positioned nose down before measuring for separation between the cast and the base. A feeler gage shall be used to measure the separation.

5. PACKAGING

5.1 Packing Level A. The projectile metal parts assy with loaded ogive shall be pallatized in accordance with dwg. E-15-12-321.

5.2 Marking. Marking shall be in accordance with dwg. E15-12-321.

6. NOTES

6.1 Intended use. The projectile metal parts assy with loaded ogive covered by this specification are intended for the Projectile, 155MM, M687.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment design for approval. See MIL-A-48078A. Submit equipment designs as required to: Commander, ARDC, ATTN: AMSMC-QAR-I, Dover, New Jersey 07801-5001.

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6.4 Data requirements. Data item descriptions required for submission under this specification are as follows:

Paragraph	Data Requirement	Applicable DID
4.5.2	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724

Two copies of the above data shall be forwarded in accordance with the requirements of the contract to: Commander, ARDC, ATTN: AMSMC-QAR-Q (1) and ATTN: DRCPM-CWA (1), Dover, New Jersey 07801-5001.

6.5 Submission of ballistic test data and data cards. In addition to the normal distribution, two copies of all ballistic data and ammunition cards shall be forwarded to: Commander, ARDC, ATTN: AMSMC-QAR-Q (1) and ATTN: DRCPM-CWA (1), Dover, New Jersey 07801-5001.

6.6 Radiographic procedures. Radiographic procedures and sample radiographs shall be submitted to: Commander, ARDC, ATTN: AMSMC-QAR-Q, Dover, New Jersey 07801-5001 for review and approval.

6.7 Photographic densities. "American Standards for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, New York, has been satisfactory as a source of information on film densities.

6.8 Clean explosive scrap. Clean explosive scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations and scrap formed during the drilling operation.

6.9 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

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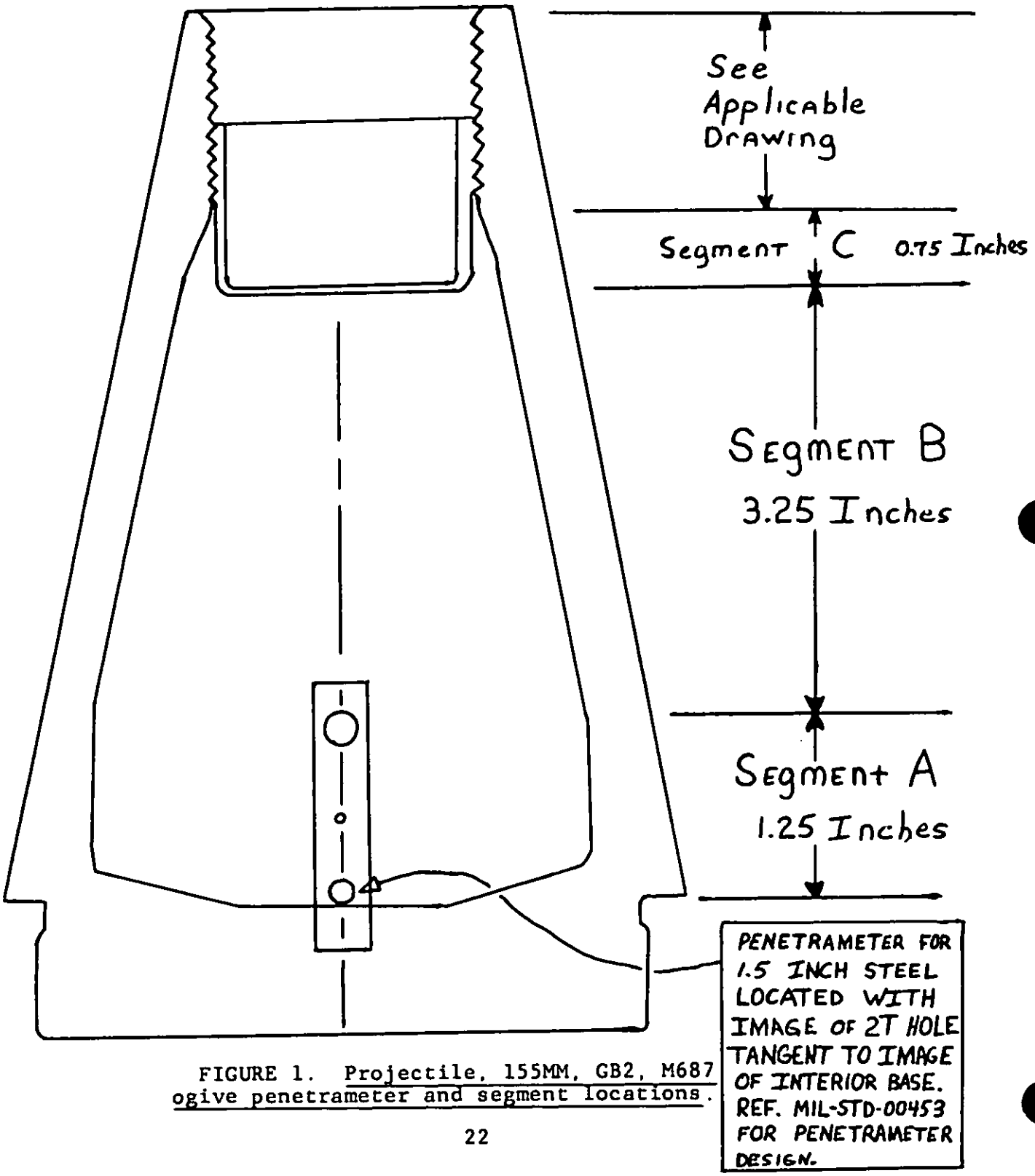


FIGURE 1. Projectile, 155MM, GB2, M687 ogive penetrameter and segment locations.

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Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1320-A671)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-64005	PROJECTILE, 155MM, GB2, M687, METAL PARTS ASSEMBLY WITH LOADED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)* OGIVE
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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